A SQL - Based Data Analytics Project

CUSTOMER RELATIONSHIP MANAGEMENT (CRM) ANALYSIS

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ABSTRACT

This report presents a structured SQL-based analysis of a Customer Relationship Management (CRM) dataset to evaluate sales performance, revenue distribution and operational efficiency across multiple business dimensions. By leveraging deal pipeline data, account information, product hierarchy and sales team structure, the analysis uncovers insights related to deal progression, sales agent productivity, revenue contribution by products and industries, regional performance, and high-value opportunities. The findings support data-driven decision-making for sales strategy optimization, resource allocation, and performance management.

ER - DIAGRAM

accounts *
account
sector
year_established
revenue
employees
office_location
subsidiary_of
sales_pipeline *
opportunity_id
sales_agent
product
account
deal_stage
engage_date
close_date
close_value
products *
product
series
sales_price
sales_teams *
sales_agent
manager
regional_office

TABLE STRUCTURE

1. Accounts Table

```
EXEC sp_help 'accounts';
```

	Column_name	Type	Computed	Length	Prec	Scale	Nullable
1	account	varchar	no	50			no
2	sector	varchar	no	50			no
3	year_established	smallint	no	2	5	0	no
4	revenue	float	no	8	53	NULL	no
5	employees	int	no	4	10	0	no
6	office_location	varchar	no	50			no
7	subsidiary_of	varchar	no	50			yes

In this table, the account column defines the name of the company, sector in which it operates & location of the office. It contains account(PK), sector, year established, revenue, no. of employees, office location and subsidiary company.

TABLE STRUCTURE

2. Products Table

```
EXEC sp_help 'products';
```

	Column_name	Type	Computed	Length	Prec	Scale	Nullable
1	product	varchar	no	50			no
2	series	varchar	no	50			no
3	sales_price	smallint	no	2	5	0	no

Products table maps the product and series names to their selling price. It contains product(PK), series and sales price.

TABLE STRUCTURE

3. Sales Teams Table

```
EXEC sp_help 'sales_teams';
```

	Column_name	Type	Computed	Length	Prec	Scale	Nullable
1	sales_agent	nvarchar	no	100			no
2	manager	nvarchar	no	100			no
3	regional_office	nvarchar	no	100			no

Sales Teams table defines each sales agents(PK), manager names and their regional office.

TABLE STRUCTURE

4. Sales Pipeline Table

```
EXEC sp_help 'sales_pipeline';
```

	Column_name	Type	Computed	Length	Prec	Scale	Nullable
1	opportunity_id	varchar	no	50			no
2	sales_agent	varchar	no	50			no
3	product	varchar	no	50			no
4	account	varchar	no	50			yes
5	deal_stage	varchar	no	50			no
6	engage_date	date	no	3	10	0	yes
7	close_date	date	no	3	10	0	yes
8	close_value	int	no	4	10	0	yes

Sales pipeline table is the fact table that connects to each of the 3 dimension tables. This table contains opportunity id, sales agent, product, account, deal stage, engage date, close date & close value.

- Deal stage represents the current status of a sales opportunity.
- Engage date represents the date when the deal first moved into the “Engaging” stage.
- Close date represents the date when the deal was finally decided either – won or lost.
- Close value is the final monetary value of the deal (revenue).

ASSUMPTIONS

- Only “Won” deals represent realized revenue because deals in other stages do not generate confirmed revenue.
- Missing close values are treated as zero where required because some deal records may not have a finalized deal value.
- Close value is populated only for closed deals because open or lost deals typically do not have finalized revenue amounts.
- Sales agent names uniquely identify individuals because no duplicate agent identifiers were present in the dataset.

Sales Pipeline Overview & Deal Distribution

Show the total number of deals in each deal stage.

```sql
SELECT deal_stage, COUNT(deal_stage) AS No_of_deals
FROM sales_pipeline
GROUP BY deal_stage;
```

	deal_stage	No_of_deals
1	Won	4238
2	Engaging	1589
3	Lost	2473
4	Prospecting	500

Insight:
Deal volume is unevenly distributed across pipeline stages, revealing where opportunities accumulate or stall.

Sales Pipeline Overview & Deal Distribution

Find the total deal value for each deal stage.

```sql
SELECT deal_stage, SUM(COALESCE(close_value,0)) AS 'total_deal_value'
FROM sales_pipeline
GROUP BY deal_stage;
```

	deal_stage	total_deal_value
1	Won	10005534
2	Engaging	0
3	Lost	0
4	Prospecting	0

Insight:
Revenue potential varies significantly by deal stage, highlighting where the highest monetary value resides within the pipeline.

Sales Pipeline Overview & Deal Distribution

Show the number of deals handled by each sales agent.

```sql
SELECT sales_agent, COUNT(deal_stage) AS No_of_deals
FROM sales_pipeline
GROUP BY sales_agent;
```

	sales_agent	No_of_deals
1	Garret Kinder	123
2	Versie Hillebrand	361
3	Jonathan Berthelot	345
4	Donn Cantrell	275
5	Corliss Cosme	310
6	Moses Frase	260
7	Maureen Marcano	285
8	Hayden Neloms	202
9	Daniell Hammack	259
10	Markita Hansen	306
11	Wilburn Farren	110
12	Kami Bicknell	362
13	Violet Mclelland	261
14	Reed Clapper	237
15	Elease Gluck	177
16	Rosalina Dieter	160
17	Niesha Huffines	239
18	Kary Hendrixson	438
19	Cecily Lampkin	203
20	Cassey Cress	346
21	Boris Faz	210
22	Anna Snelling	448
23	Vicki Laflamme	451
24	Zane Levy	349
25	Lajuana Vencill	311
26	Gladys Colclough	317
27	James Ascencio	267
28	Marty Freudenburg	281
29	Rosie Papadopo...	160
30	Darcel Schlecht	747

Insight:
Deal ownership differs widely among sales agents, indicating variation in workload and sales coverage.

Sales Pipeline Overview & Deal Distribution

Display sales agents who have handled more than 350 deals in descending order.

```sql
SELECT sales_agent, COUNT(deal_stage) AS No_of_deals
FROM sales_pipeline
GROUP BY sales_agent
HAVING COUNT(deal_stage)>350
ORDER BY No_of_deals DESC;
```

	sales_agent	No_of_deals
1	Darcel Schlecht	747
2	Vicki Laflamme	451
3	Anna Snelling	448
4	Kary Hendrixson	438
5	Kami Bicknell	362
6	Versie Hillebrand	361

Insight:
A small group of agents manages a disproportionately high number of deals, suggesting reliance on high-capacity performers.

Sales Agent Performance Analysis

List the top 5 sales agents by total revenue from “Won” deals.

```sql
SELECT TOP 5 sales_agent, SUM(COALESCE(close_value,0)) AS total_revenue
FROM sales_pipeline
WHERE deal_stage = 'Won'
GROUP BY sales_agent
ORDER BY total_revenue DESC;
```

	sales_agent	total_revenue
1	Darcel Schlecht	1153214
2	Vicki Laflamme	478396
3	Kary Hendrixson	454298
4	Cassey Cress	450489
5	Donn Cantrell	445860

Insight:
A limited set of sales agents contributes the majority of realized revenue, underscoring their critical role in business growth.

Sales Agent Performance Analysis

Find the 5 most recent closed deals with their deal value and the name of the sales agent.

```
SELECT TOP 5 close_date,sales_agent, close_value
FROM sales_pipeline
WHERE close_date is NOT NULL
ORDER BY close_value DESC
```

	close_date	sales_agent	close_value
1	2017-06-07	Rosalina Dieter	30288
2	2017-06-27	Markita Hansen	29617
3	2017-12-04	Elease Gluck	29220
4	2017-09-26	Markita Hansen	29166
5	2017-10-29	Elease Gluck	27971

Insight:
Recent deal closures reflect ongoing sales momentum and provide a snapshot of current revenue inflow.

Sales Agent Performance Analysis

Display sales agents who have closed at least one deal.

```
SELECT DISTINCT sales_agent
FROM sales_pipeline
where close_value >= 1;
```

	sales_agent
1	Garret Kinder
2	Versie Hillebrand
3	Jonathan Berthelot
4	Donn Cantrell
5	Corliss Cosme
6	Moses Frase
7	Maureen Marcano
8	Hayden Neloms
9	Daniell Hammack
10	Markita Hansen
11	Wilburn Farren
12	Kami Bicknell
13	Violet Mclelland
14	Reed Clapper
15	Elease Gluck
16	Rosalina Dieter
17	Niesha Huffines
18	Kary Hendrixson
19	Cecily Lampkin
20	Cassey Cress
21	Boris Faz
22	Anna Snelling
23	Vicki Laflamme
24	Zane Levy
25	Lajuana Vencill
26	Gladys Colclough
27	James Ascencio
28	Marty Freudenburg
29	Rosie Papadopo...
30	Darcel Schlecht

Insight:
Only 30 out of 35 sales agents actively converts opportunities into revenue, highlighting gaps in productivity.

Revenue & Product Performance

Calculate total revenue generated by each product series.

```sql
SELECT p.series, SUM(sp.close_value) AS total_revenue FROM products p
LEFT JOIN sales_pipeline sp
ON sp.product = p.product
WHERE close_value IS NOT NULL
GROUP BY series;
```

	series	total_revenue
1	GTK	400612
2	GTX	3834189
3	MG	2260155

Insight:
Revenue concentration varies across product series, revealing which offerings drive the most financial impact.

Revenue & Product Performance

Calculate total revenue generated by each industry (sector) and product series.

```sql
SELECT a.sector, p.series, SUM(sp.close_value) AS total_revenue FROM accounts a
INNER JOIN sales_pipeline sp
ON a.account = sp.account
INNER JOIN products p
ON p.product = sp.product
WHERE sp.deal_stage = 'Won'
GROUP BY a.sector, p.series
ORDER BY total_revenue DESC;
```

	sector	series	total_revenue
1	retail	GTX	735814
2	technolgy	GTX	583347
3	medical	GTX	554363
4	retail	MG	419042
5	software	GTX	393444
6	marketing	GTX	375649
7	technolgy	MG	350114
8	finance	GTX	336222
9	medical	MG	321680
10	telecommunications	GTX	274232
11	software	MG	244442
12	services	GTX	222009
13	entertainment	GTX	219826
14	finance	MG	214717
15	marketing	MG	186368
16	entertainment	MG	179013
17	telecommunications	MG	149480
18	employment	GTX	139283
19	entertainment	GTK	108498
20	employment	MG	103309
21	services	MG	91990
22	finance	GTK	55406
23	retail	GTK	55237
24	medical	GTK	51131
25	employment	GTK	50033
26	marketing	GTK	29617
27	software	GTK	25791
28	telecommunications	GTK	24899

Insight:
Specific product-industry combinations generate significantly, higher revenue, indicating strong market-product alignment.

Industry & Regional Insight

Retrieve distinct industries (sectors) along with the number of companies in each sector.

```sql
SELECT sector, COUNT(account) AS No_of_companies
FROM accounts
GROUP BY sector
ORDER BY No_of_companies DESC;
```

	sector	No_of_companies
1	retail	17
2	medical	12
3	technolgy	12
4	finance	8
5	marketing	8
6	software	7
7	entertainment	6
8	telecommunications	6
9	services	5
10	employment	4

Insight:
Certain industries dominate the customer base, indicating stronger market penetration in specific sectors.

Industry & Regional Insight

Show total revenue generated by each regional office.

```sql
SELECT s.regional_office, SUM(sp.close_value) AS total_revenue FROM sales_teams s
INNER JOIN sales_pipeline sp
ON s.sales_agent = sp.sales_agent
WHERE sp.deal_stage = 'Won'
GROUP BY s.regional_office;
```

	regional_office	total_revenue
1	Central	3346293
2	East	3090594
3	West	3568647

Insight:
Revenue contribution is relatively evenly distributed across regional offices, indicating consistent sales performance with the West region showing a slight lead.

Industry & Regional Insight

Identify the top 3 accounts by total revenue, along with their industry and assigned sales manager.

```sql
SELECT TOP 3 a.account, a.sector, s.manager, SUM(sp.close_value) AS total_revenue
FROM accounts a
LEFT JOIN sales_pipeline sp
ON a.account = sp.account
LEFT JOIN sales_teams s
ON s.sales_agent = sp.sales_agent
WHERE sp.deal_stage = 'Won'
GROUP BY a.account, a.sector, s.manager
ORDER BY total_revenue DESC;
```

	account	sector	manager	total_revenue
1	Treequote	telecommunications	Melvin Marxen	133833
2	Cheers	entertainment	Celia Rouche	122963
3	Kan-code	software	Rocco Neubert	119566

Insight:
The top-revenue accounts are distributed across telecommunications, entertainment and software sectors, indicating that high-value customers span multiple industries rather than being concentrated in a single sector. Expanding similar high-value accounts within these sectors can help scale revenue while maintaining a balanced industry portfolio.

Industry & Regional Insight

Count the number of deals Won by each office location within first 15 days of March.

```sql
SELECT a.office_location, COUNT(sp.deal_stage) AS deals_won FROM accounts a
INNER JOIN sales_pipeline sp
ON a.account = sp.account
WHERE sp.deal_stage = 'Won'
AND sp.close_date BETWEEN '2017-04-01' AND '2017-04-15'
GROUP BY a.office_location
ORDER BY deals_won DESC;
```

	office_location	deals_won
1	United States	113
2	Korea	5
3	Jordan	3
4	Kenya	3
5	Philipines	3
6	Poland	3
7	Japan	2
8	Norway	2
9	Panama	2
10	Belgium	1
11	Brazil	1
12	Germany	1
13	Italy	1

Insight:
Deal closures during the selected time period are heavily concentrated in the United States, while other office locations show minimal activity, indicating a strong short-term dependency on a single geography.
Strengthening sales execution and pipeline conversion in underperforming locations could reduce geographic dependency and improve overall resilience.

Advanced Analytics & Benchmarking

Show the top 10 sales agents along with their total revenue and revenue rank.

```sql
WITH x AS
(SELECT sales_agent, SUM(close_value) AS total_revenue,
RANK() OVER (ORDER BY SUM(close_value) DESC) AS revenue_rank
FROM sales_pipeline
WHERE deal_stage = 'Won'
GROUP BY sales_agent)
SELECT sales_agent, total_revenue, revenue_rank FROM x
WHERE revenue_rank <= 10;
```

	sales_agent	total_revenue	revenue_rank
1	Darcel Schlecht	1153214	1
2	Vicki Laflamme	478396	2
3	Kary Hendrixson	454298	3
4	Cassey Cress	450489	4
5	Donn Cantrell	445860	5
6	Reed Clapper	438336	6
7	Zane Levy	430068	7
8	Corliss Cosme	421036	8
9	James Ascencio	413533	9
10	Daniell Hammack	364229	10

Insight:
Revenue contribution among the top sales agents is uneven, with the top-ranked agent generating a significantly higher revenue than the rest, indicating strong performance concentration at the top.

Advanced Analytics & Benchmarking

Find products whose average deal value is higher than the overall average.

```sql
SELECT product, AVG(close_value) AS avg_deal_value FROM sales_pipeline
WHERE deal_stage = 'Won'
GROUP BY product
HAVING AVG(close_value) > (SELECT AVG(close_value) FROM sales_pipeline);
```

	product	avg_deal_value
1	GTK 500	26707
2	GTX Plus Pro	5489
3	GTXPro	4815
4	MG Advanced	3388

Insight:
Certain products consistently achieve higher-than-average deal values, indicating stronger pricing power or higher customer demand.

Advanced Analytics & Benchmarking

List sales agents who closed deals worth more than the average closed deal value.

```
SELECT DISTINCT sales_agent FROM sales_pipeline
WHERE deal_stage = 'Won' AND
close_value > (SELECT AVG(close_value) FROM sales_pipeline
WHERE deal_stage = 'Won');
```

	sales_agent
1	Garret Kinder
2	Versie Hillebrand
3	Jonathan Berthelot
4	Donn Cantrell
5	Corliss Cosme
6	Moses Frase
7	Maureen Marcano
8	Hayden Neloms
9	Daniell Hammack
10	Markita Hansen
11	Wilburn Farren
12	Kami Bicknell
13	Violet Mclelland
14	Reed Clapper
15	Elease Gluck
16	Rosalina Dieter
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27	James Ascencio
28	Marty Freudenburg
29	Rosie Papadopo...
30	Darcel Schlecht

Insight:
Some sales agents consistently close higher-value deals, reflecting stronger negotiation capability or focus on strategic opportunities.

Conclusion

- This CRM analysis highlights clear patterns in how deals progress, where revenue is concentrated and which sales agents, products, industries, and regions contribute most to business outcomes.
- Revenue generation is driven by a relatively small set of high-performing agents, key accounts and specific product-industry combinations, while performance varies noticeably across regions and sales personnel.
- These insights indicate opportunities to improve overall performance by redistributing workloads, replicating best practices from top performers, strengthening underperforming regions and prioritizing high-value products and customers.
- By acting on these findings, the organization can enhance sales efficiency, improve deal conversion quality and drive more consistent and scalable revenue growth.